SectraTEK

AGENTIC AI FOR DIGITAL FORENSICS · LAB 02

# Giving AI Eyes on the Evidence



The Tool Use Pattern: instead of guessing from a prompt, the model calls real tools that read the artifacts, then reasons over what they return.

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YOUR INSTRUCTOR

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IN ONE LINE

I study how phones get spied on and tampered with, and build AI tools that help examiners find the evidence.

**Mobile threat research**
Spyware, stalkerware and device tampering.

**AI for investigators**
Machine learning that speeds up the search.

**Teaching**
AI for Cybersecurity and Digital Forensics.

TODAY'S STORY

# From self-review to real evidence

1

## What tools add

The model stops guessing and starts inspecting.

2

## Before AI

Examiners opened each artifact by hand.

3

## The problem

A model with no access invents the details.

4

## The discipline

Ground every claim in a tool output.

5

## The gap

Reasoning without access is still guessing.

6

## Tool Use

The model requests a tool; the tool reads the data.

7

## The case

A stolen SUV, a phone, and three tools.

8

## Your lab

Call the tools, then let the agent call them.

Practical runs in Google Colab with a free hosted model. Nothing is installed on your computer.

WHERE WE ARE

# Lab 01 fixed the words. Lab 02 fixes the eyes.

## Lab 01 · Reflection

The model reviewed its own draft and pulled overclaims back to the evidence. It improved how it wrote.

But it still only saw what was pasted into the prompt.

TODAY

## Lab 02 · Tool Use

Now the model can call tools that open the real artifacts: the gallery, the metadata, the sale drafts.

It stops guessing about the evidence and starts reading it.

THE KEY IDEA

# A tool is a function the model can call

ONE TOOL, THREE PARTS

**A name**
inspect_image_evidence

**Typed inputs**
file_name, case_description

**An output**
the record the model reads

WHY THIS MATTERS

The model never touches the phone data directly. It asks a tool, by name, with valid inputs. The tool reads the evidence and hands back a result the model can reason over. That keeps the model honest: its answer is built on a real observation, not a guess.

IN THIS LAB

list_media_files finds photos  inspect_image_evidence reads one photo  inspect_listing_records reads sale drafts

THE MANUAL CRAFT

# How an examiner inspects this phone

### Open the gallery

Scroll the DCIM folder for photos near the theft time.

### Read the metadata

Check each photo's capture time, GPS, and camera.

### Check detections

Confirm the color, body type, and roof rack.

### Open the drafts

See which photo is attached to a sale listing.

**RELIABLE, BUT SLOW AND SERIAL**

An examiner does this one artifact at a time, cross-checking by hand. It is accurate, but it does not keep up when there are thousands of photos and a model that could help, if only it could reach the data.

THE PROBLEM

# A model with no access invents the details

ASK A PLAIN MODEL, WITH NO TOOLS

“IMG_3310.jpg, taken at 19:42, clearly shows the stolen SUV, and the listing was posted to Marketplace at 20:05.”

EVERY SPECIFIC IS FABRICATED

### A filename that does not exist

The real candidate is IMG_2044.jpg. The model guessed.

### A timestamp from nowhere

No artifact says 19:42. It was invented to sound precise.

### A sale that never happened

No draft was ever published. “Posted” is pure hallucination.

THE DISCIPLINE

# Do not ask the model. Ask the tool.

## Ask the model what the photo shows

- The model guesses from the prompt.
- It invents filenames and times.
- The answer cannot be checked.

## Ask a tool, then let the model reason

- The tool reads the real record.
- The model reasons over the result.
- Every claim traces to an output.

Same model, same case. The difference is whether a real observation sits under each claim.

THE GAP

# The model can reason, but it cannot see

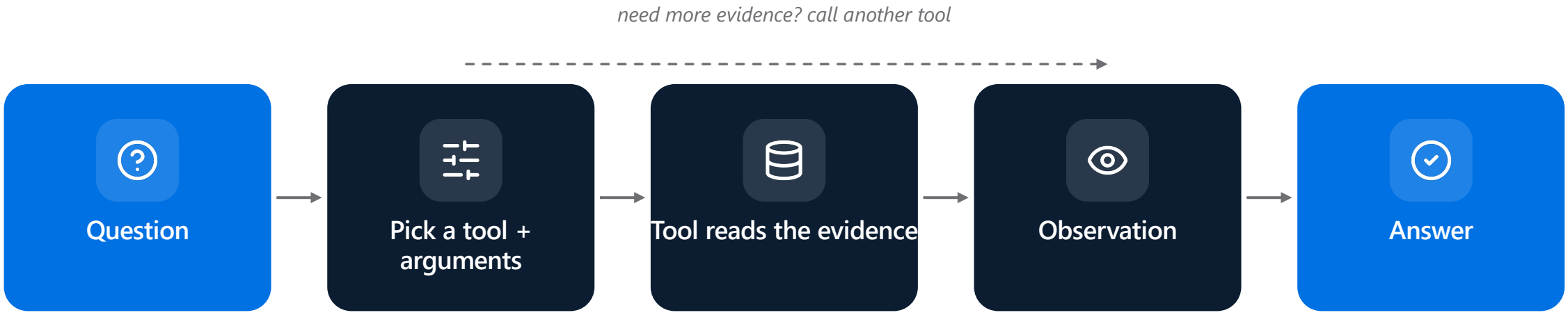



**INTELLIGENCE IS NOT ACCESS**

The model can weigh evidence as well as a junior analyst, but it has no hands and no eyes on the device. Reasoning over evidence it cannot read is just confident guessing. Tool Use is the bridge across this gap: it gives the model a safe, named way to reach the data.

THE ANSWER

# The Tool Use Pattern




**THE MODEL NEVER GUESSES THE DATA**

It only decides which tool to call and with what arguments. The tool returns the facts.

**WHEN DOES IT STOP?**

When the model has enough observations to answer, it writes the report instead of calling another tool.

UNDER THE HOOD

# How the model asks for a tool

THE MODEL WRITES A REQUEST

```
<tool_call>
{"name":
   "inspect_image_evidence",
 "arguments": {
   "file_name":
     "IMG_2044.jpg",
   "case_description":
     "black SUV with
      roof rack"}}
</tool_call>
```

### The agent reads the request

It pulls the tool name and arguments out of the block.

### It checks the arguments

Each value is coerced to the type the tool declared.

### It runs the real function

The tool reads the evidence and returns a result.

### The result goes back

The model reasons over it, or asks for another tool.

THE INVESTIGATION

# Case TU-2026-004

THE DEVICE

PHONE

Samsung Galaxy S23, Android 14

ACQUIRED

Magnet AXIOM, logical + media

REPORTED STOLEN

black SUV with roof rack

RECOVERED

a phone used by the seller

THE QUESTION

Does the phone show that the stolen SUV was photographed and prepared for an online sale?

**A photo**
IMG_2044.jpg: a black SUV with a roof rack, captured 21:14 UTC.

**A draft**
A listing titled “black SUV for sale” with that photo attached, 21:31 UTC.

**A boundary**
The draft was never published. Preparation is not a completed sale.

YOUR INSTRUMENTS

# Three tools read the evidence

### list_media_files

*root, date_from*

Finds candidate photos in a folder, on or after a date.

### inspect_image_evidence

*file_name, case_description*

Combines one photo's metadata and vehicle detection, and scores the match.

### inspect_listing_records

*date_from*

Returns saved sale drafts and which photo each one attaches.

Core sequence: **list_media_files** -> **inspect_image_evidence** -> **inspect_listing_records**

SEE THE TOOLS BUILD A CONCLUSION

# Three calls, one careful answer

| | | |
|---|---|---|
| 1 | **list_media_files("DCIM/Camera", from 2026-01-02)**<br>-> 3 candidate photos | Narrows review to photos taken on or after the theft date. |
| 2 | **inspect_image_evidence("IMG_2044.jpg", "black SUV with roof rack")**<br>-> strong match, 21:14 UTC | Black, SUV, and roof rack all agree with the stolen vehicle. |
| 3 | **inspect_listing_records(from 2026-01-02)**<br>-> "black SUV for sale", IMG_2044 attached, 21:31 UTC | Links the same photo to a sale draft created minutes later. |

**Defensible conclusion:** confirmed evidence the SUV was photographed and prepared for sale. The draft was never published, so a completed sale is not confirmed.

SAY EXACTLY HOW SURE YOU ARE

# Confirmed, likely, or unconfirmed

## Confirmed

Direct tool outputs support each part of the claim.

**EXAMPLE**

The SUV was photographed and attached to a sale draft.

## Likely

The outputs suggest it, but one link is missing or weak.

**EXAMPLE**

A second photo matches, but its roof rack was not detected.

## Unconfirmed

The outputs do not adequately support the claim.

**EXAMPLE**

That the listing was ever posted, or the vehicle sold.

PRACTICAL

# Your turn, in the browser

| | | |
|---|---|---|
| 1 | **Open in Colab** | Click the badge in the notebook. Nothing to install. |
| 2 | **Paste a free key** | Get a free Groq key, paste it when asked. It is hidden. |
| 3 | **Call the tools yourself** | Part A: run the three tools and read every result. |
| 4 | **Hand them to ToolAgent** | Part B: watch the agent pick the calls, then judge them. |
| 5 | **Write your report** | Label the evidence confirmed, likely, or unconfirmed. |

SectraTEK

# A model is only as good as what it can check.

Next lab: ReAct, where the agent reasons one step, acts, observes, and repeats until the answer holds.

